Spike: 6
Title: Emergent Group Behaviour

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Goals / deliverables:
The aim of this spike is to demonstrate a working simulation with different modes of emergent group behaviour. This includes the implementation of cohesion, separation, alignment and wandering steering behaviours as well as a user interface which allows users to both see and change these parameters in real-time.

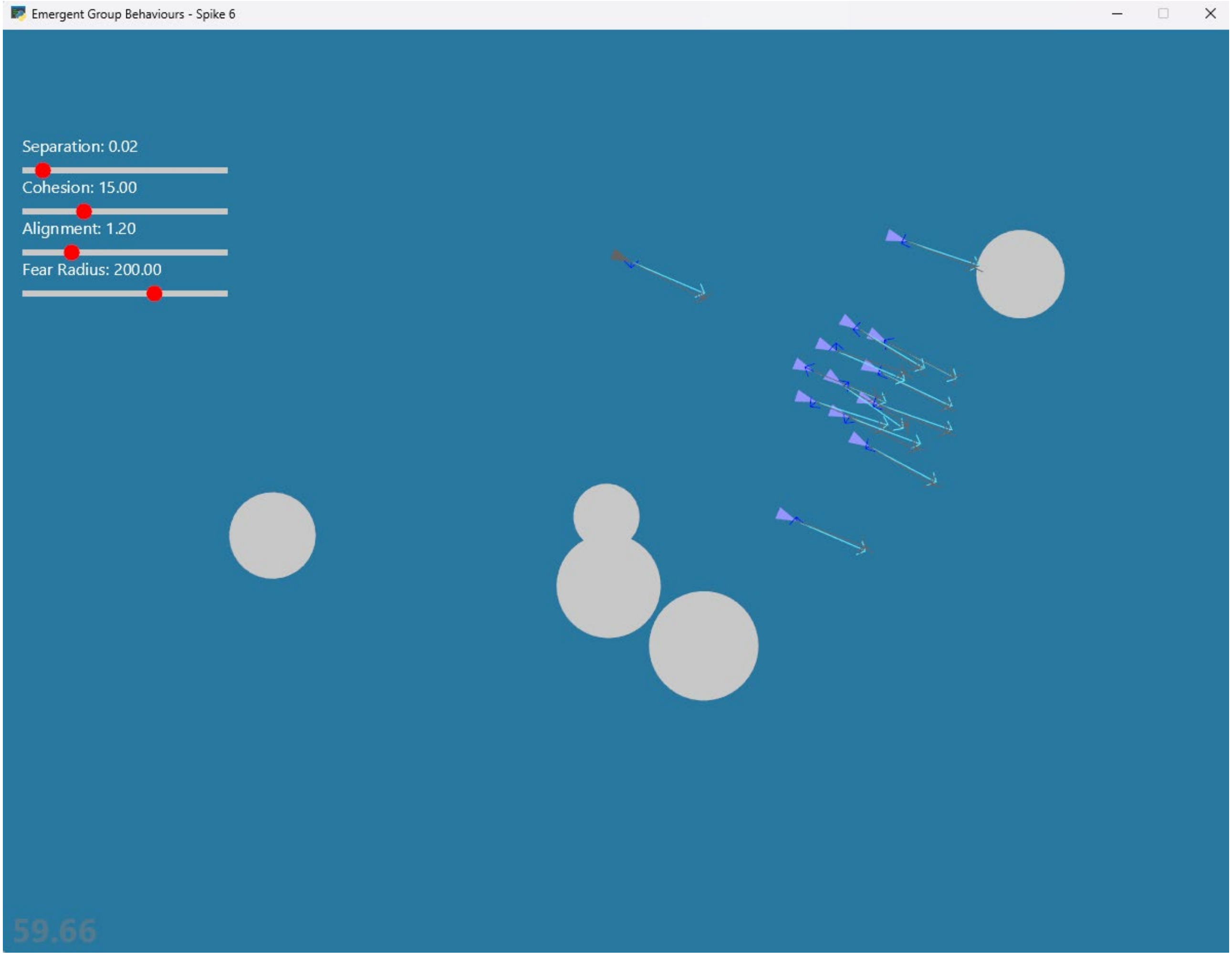
Technologies, Tools, and Resources used:

List of information needed by someone trying to reproduce this work:
- Visual Studio Code
- Python 3.x
- Pyglet 2.1.x
- Git
- Github (for installation of code)

Tasks undertaken:
- Download and install Visual Studio Code
- Download and install the latest version of Python
- Install the latest version of Pyglet (pip install --upgrade --user piglet)
- Import the lab containing the base code for this spike (Task 13)
- Implement grouping behaviour and different parameters
- Implement a simple UI which allows users to change these parameters
- Create a predator and prey fleeing behaviour
- Add all required debugging information

What we found out:

Figure 1: Two groups emerge. The grey agent denotes the hunter.



A single contiguous school of prey flee from the hunter as they are within its fear radius. Extra prey agents have been added. The separation value is kept low, as it uses an inverse square function to derive its strength (similarly to gravity, but in the opposite direction).